25900519

MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Discipline Specific Core Course (DSC) - MG3DSCMBG202 - TECHNIQUES IN MICROBIOLOGY
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions: Answer any **10** questions

Each question carries **2 marks**

1. Define sanitisation. [K] / [CO1]
   The process of reducing the microbial population to a safe level as determined by public health standards. (2)
2. Define simple media. [K] / [CO2]
   Simple media are general-purpose media that support most non-fastidious bacteria. (1)Eg: Peptone water, nutrient broth, nutrient agar (1)
3. Mention two tools used for soil sample collection. [K] / [CO3]
   Spade or auger, Core sampler (2)
4. What is the primary purpose of staining in microbiology? [K] / [CO4]
   The primary purpose of staining is to make microscopic and mostly colourless bacteria more visible, and to study their structure, shape, and other characteristics. (2)
5. Name one cryoprotectant commonly used in cryopreservation. [K] / [CO3]
   Dimethyl sulfoxide (DMSO), Glycerol, Sucrose (2)
6. What is the mode of action of aldehydes in sterilisation? [K] / [CO1]
   They cause alkylation of amino-, carboxyl- or hydroxyl group, and probably damages nucleic acids. They are sporicidal. (2)
7. Name two indicator media. [K] / [CO2]
   MacConkey agar, Eosin Methylene Blue (EMB) Agar, Tellurite Blood Agar (TBA) (2)
8. Give an example of a long-term microbial preservation method. [K] / [CO3]
   Any one method- Lyophilisation or freeze-drying, Cryopreservation (2)
9. Name the mordant used in Gram staining. [K] / [CO4]
   Grams iodine(2)
10. Give a note on the candle jar method. [K] / [CO2]
    The candle jar method is a simple technique used in microbiology to create an anaerobic condition for culturing anaerobic bacteria. (1) Most suitable for the growth of

microaerophilic and capnophilic bacteria. Simple, inexpensive, and does not require special equipment. But Oxygen concentration cannot be precisely controlled. (1)

11. What is BSC? [K] / [CO1]
    Biological safety cabinets (1). Enclosed, ventilated laboratory workspace for safely working with materials with pathogens requiring a defined biosafety level. (1)
12. Name the oxygen indicator used in thioglycolate broth. [K] / [CO2]
    Methylene blue (2)

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5 marks**

13. Describe the method of preservation of microbial cultures by refrigeration. [U] / [CO3]
    Pure cultures can be successfully stored at 4^0 °C in refrigerators. This method is a short-term preservation method. 2-3 weeks for bacteria and 3-4 months for fungi. The metabolic activity of the microorganisms is greatly slowed down but not stopped. Their growth continues slowly as nutrients are utilised and waste products are released into the medium finally results in the death of the microbes after some time.
14. Explain in detail the steps involved in the wet mount technique. [U] / [CO4]
    The wet mount technique is a simple and common method used to view live microbes under a light microscope. It allows for the study of motility, shape, size, and arrangement of microbes in their natural state without staining or killing them. Explain - Slide preparation, Specimen inoculation, use of coverslip, Observation under microscope.
15. Describe the sterilisation controls used to test the effectiveness of a hot air oven. [U] / [CO1]
    Physical, chemical, and biological methods.
    Physical Controls: monitoring and verifying the operating parameters of the oven – Temperature and time by using thermometers
    Chemical Indicators: Explain about Browne's Tube and Chemical Indicator Strips or Tapes.
    Biological Indicators: The spores of a non-toxigenic strain of *Clostridium tetani.*
16. Explain the principle of the oil-immersion objective in a compound microscope. [U] / [CO4]

    An oil immersion objective increases the resolving power of a microscope. When light passes from the glass slide (higher refractive index) to air (lower refractive index) and then into the objective lens (higher refractive index), some light is bent away and lost, reducing image clarity and brightness. By placing a drop of immersion oil (which has a refractive index similar to that of glass, around 1.5), light rays can pass straight into the objective without significant refraction. More light enters the lens, resulting in a brighter, clearer, and higher-resolution image.

17. Write a short note on enriched media with examples. [K] / [CO2]
    Enriched media are basal media (like nutrient agar or broth) supplemented with extra nutrients (e.g., blood, serum, or growth factors) to promote the growth of fastidious

microorganisms that do not grow well on ordinary media. E.g.: Blood agar, chocolate agar, Loeffler's serum slope, etc.

18. Explain the principle, procedure and types of the streak plate method. [U] / [CO3]
The most common technique of isolation of bacterial cells on the surface of an agar plate is the streaking method. The key principle of this method is that, by streaking, a dilution gradient is established across the surface of the petri plate as bacterial cells are deposited on the agar surface. Each colony is the progeny of a single microbial cell, thus
representing a clone of pure culture.
Principle – (3 marks)
Procedure – (3 marks)
Different methods can be used for streaking. Common ones are the quadrant (four-way streaks), continuous streak and radiant streak methods. (4 marks)

Part C

Long Essay Questions

Answer any **1** question

Each question carries **10 marks**

19. Describe the different types of radiation and filtration techniques used in sterilisation. [U] / [CO1]

Filtration: Helps to remove bacteria from heat-labile liquids such as serum, solution of sugars or antibiotics. Explain the different types of filtrations - Candle filter, Asbestos filter, Sintered glass filter, Membrane filter. (5 marks)
Radiations: Explain the two types. (5 marks).
Ionising radiation: e.g., X-rays, gamma rays, and high-speed electrons.
Non-ionising radiation: e.g. ultraviolet light and infrared light.

20. Compare the spread plate and pour plate techniques with neat diagrams. [U] / [CO3]
Explain the pour plate with a diagram (5 marks) and the spread plate with a diagram (5 marks).